

NAME:

ADM NO:

DEPARTMENT:

SECTION:

SEMESTER: WINTER, SESSION: 2024-2025

Subject: Digital Electronics [Quiz-1]
Time: 30 Mins

Subject Code: NECE102
Maximum Marks: 10

Instructions: For questions 1 to 5, tick the correct option, and for 6 to 10, fill in the blanks.

1. Simplified Expression for the following k-map is_____

- a) $B'D' + A'B'$
b) $BD + C'D$
c) $B'D' + C'D'$
d) $C'D' + A'B'D'$

AB \ CD	00	01	11	10
00	1			1
01	1			
11	1			X
10	1			X

2. If $F(w, x, y, z) = \sum m(1, 3, 5, 7, 9, 11, 13, 14, 15)$ then the simplified expression for “F” is
(a) $x'z' + y'z'$ (b) $x'z + y'z'$ (c) $x + yz$ (d) $xz + yz'$ **(e) None**

3. The simplified expression for the function $Y(x, y, z) = \sum m(0, 2, 4, 6)$ is

- (a) x' (b) y' **(c) z'** (d) $x'y'$ (e) None

4. $F = AB + CD + B\bar{D}$. Express F in POS form.

- a) $(\bar{A} + \bar{B})(\bar{C} + \bar{D})(\bar{B} + D)$ c) $(A + B)(C + D)(B + \bar{D})$
b) $(B + D)(B + C)(A + C + \bar{D})$ d) $(\bar{B} + \bar{D})(\bar{B} + \bar{C})(\bar{A} + \bar{C} + D)$

5. What is the minimum number of gates required to implement the Boolean Function $(AB+C)$ with only 2-input NOR gates?

- (a) 4 (b) 5 **(c) 3** (d) 6 (e) None

6. Find $(756.603)_8 = (\underline{\hspace{2cm}} \text{1EE.C18} \hspace{2cm})_{16}$

7. Find $(1529.75)_{16} = (\underline{\hspace{2cm}} \text{12451.352} \hspace{2cm})_8$

8. Find 2's complement of $(1001101)_2 = (\underline{\hspace{2cm}} \text{110011} \hspace{2cm})_2$

9. Minimum number of NAND gates required to realize X-NOR is (5).

10. Minimum number of NOR gates required to implement $F = \overline{A + \bar{A}B}$ is (1).